

LESSON 6-2 • HOMEWORK PACKET • SHEET 1 OF 2**Dividing a Fraction by a Fraction**

Maryland Standard 6.NS.A.1 • Learning target – I can divide a fraction by another fraction using the reciprocal.

REMEMBER

- To divide by a fraction, **multiply by its reciprocal**:
 $3/4 \div 3/8 = 3/4 \times 8/3$.
- Keep the first fraction, change $\div$ to $\times$, and flip only the **second** fraction.
- Check: quotient $\times$ divisor = dividend. $2 \times 3/8 = 6/8 = 3/4$.

WORKED EXAMPLEHow many $3/8$ -cup servings are in $3/4$ cup of yogurt?**Step 1** Write the division: $3/4 \div 3/8$ **Step 2** Multiply by the reciprocal: $3/4 \times 8/3 = 24/12$ **Step 3** Simplify: $24/12 = 2$ **Answer 2 servings****PRACTICE***show your work • label units*

1. The bar is 1 whole. The shaded part is $4/5$ of the bar. How many $1/10$ pieces fit in $4/5$? Write the division and the quotient.

**Division** _____

2. Divide. Write each quotient in simplest form.

a. $3/4 \div 1/8 =$ _____

b. $2/3 \div 2/9 =$ _____

c. $5/6 \div 1/3 =$ _____

d. $7/10 \div 2/5 =$ _____

e. $1/2 \div 3/4 =$ _____

f. $8/9 \div 4/9 =$ _____

3. **Pancakes.** A pancake recipe needs $2/3$ cup of milk. Rosa has only $1/2$ cup of milk. What fraction of the recipe can she make?

*Show your work.***Answer** _____ of the recipe

4. **Bookmarks.** Jae has $5/6$ yard of ribbon. Each bookmark needs $5/12$ yard. How many bookmarks can Jae make?

Answer _____ bookmarks

5. **Find the Error.** Imani wrote $4/5 \div 2/3 = 4/5 \times 2/3 = 8/15$. Explain her mistake. Then find the correct quotient.

CHECK YOURSELF☐ I can do this on my own☐ I need a little help☐ I have a question about #_____**AT HOME** With a measuring cup, show someone at home how many $1/4$ -cup scoops are in $3/4$ cup.

LESSON 6-2 • HOMEWORK PACKET • SHEET 2 OF 2**Dividing with Mixed Numbers**

Maryland Standard 6.NS.A.1 • Learning target – I can divide mixed numbers by writing them as improper fractions first.

REMEMBER

- Write each mixed number as an **improper fraction**:
 $2\frac{3}{4} = 11/4$.
- Then multiply by the **reciprocal** of the divisor.
- Simplify. Write an improper answer as a mixed number.

WORKED EXAMPLEA board is $5\frac{1}{4}$ feet long. How many $\frac{3}{4}$ -foot pieces can be cut?**Step 1** $5\frac{1}{4} = 21/4$, so $21/4 \div 3/4 = 21/4 \times 4/3$ **Step 2** Multiply and simplify: $84/12 = 7$ **Answer 7 pieces****PRACTICE***show your work • label units*

1. Divide. Write each quotient in simplest form.

a. $4\frac{1}{2} \div 3/4 = \underline{\hspace{2cm}}$ b. $3\frac{1}{3} \div 5/6 = \underline{\hspace{2cm}}$ c. $2\frac{1}{4} \div 1\frac{1}{2} = \underline{\hspace{2cm}}$

d. $6 \div 1\frac{1}{3} = \underline{\hspace{2cm}}$ e. $5\frac{5}{6} \div 1\frac{1}{6} = \underline{\hspace{2cm}}$ f. $1\frac{7}{8} \div 2\frac{1}{2} = \underline{\hspace{2cm}}$

2. Write each mixed number as an improper fraction.

a. $3\frac{2}{5} = \underline{\hspace{2cm}}$ b. $4\frac{5}{6} = \underline{\hspace{2cm}}$

- 3.
- Soup Bowls.**
- A pot holds
- $8\frac{3}{4}$
- cups of soup. Each bowl holds
- $1\frac{1}{2}$
- cups. How many bowls can be filled completely? How much soup is left over?

*Show your work.***Answer** _____

- 4.
- Hiking.**
- A trail is
- $7\frac{1}{2}$
- miles long. Kenji hikes
- $1\frac{1}{4}$
- miles each hour. How many hours will it take him to hike the whole trail?

Answer _____ hours

- 5.
- Reason.**
- Without dividing, decide: is
- $6\frac{1}{2} \div 1\frac{1}{4}$
- greater than or less than
- $6\frac{1}{2}$
- ?

☐ greater than $6\frac{1}{2}$ ☐ less than $6\frac{1}{2}$ **TEST PRACTICE***choose the best answer*

6. What is
- $3\frac{1}{2} \div 1\frac{3}{4}$
- ?

A $1/2$ **B** 2**C** $3\frac{2}{3}$ **D** $6\frac{1}{8}$

7. A rope is
- $6\frac{2}{3}$
- feet long. It is cut into pieces that are each
- $\frac{5}{6}$
- foot long. How many pieces are there?

A $1/8$ **B** $5\frac{5}{9}$ **C** 8**D** 12**CHECK YOURSELF**☐ I can do this on my own☐ I need a little help☐ I have a question about #_____**AT HOME** Measure $1\frac{1}{2}$ cups of water with a $\frac{1}{4}$ -cup scoop. Does the number of scoops match $1\frac{1}{2} \div \frac{1}{4}$?